

Chemistry for WA 1 TEST 1

Unit 2A, Atomic Structure and Bonding

Name: _____ Class: _____ Date: _____

Instructions: Write answers in the right-hand column.

Score: _____ / 47 marks

Section A—Multiple choice (10 marks)

1	The isotope of cobalt used to irradiate food to destroy disease-causing bacteria is $^{60}_{27}\text{Co}$. The number of electrons in a Co^{2+} ion of this isotope is: A 25 B 27 C 29 D 33.	A	1
2	The electron structure of P^{3-} is: A 2, 8, 5 B 2, 5 C 2, 8, 8 D 2, 8, 2.	C	1
3	Boron, aluminium, gallium (Ga) and indium (In) occur together in the same group in the Periodic Table because: A atoms of each of the elements contain the same number of electrons B they are good conductors of electricity C neutral atoms of each of the elements have the same number of electrons in their outer shell D the masses of the atoms of these elements increase smoothly down the group.	C	1
4	The compound formed between a group 15 element and a group 17 element would be expected to: A have a very high boiling point B possibly be a gas or a liquid at room temperature C conduct electricity in solid state D conduct electricity in molten state.	B	1
5	Which of the following molecules has two lone pair of electrons? A CH_4 B HCl C H_2O D NH_3	C	1
6	Calcium atoms and calcium ions: A are chemically identical B have the same number of electrons C have the same number of protons D both react readily with water.	C	1

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7	Which one of the following contains a triple covalent bond? A H ₂ S B NO ₃ ⁻ C HCN D C ₂ H ₄	C	1
8	Element Y has an atomic mass of 25, and a nucleus that contains 12 neutrons. Its chloride is most likely to have the formula: A Y ₂ Cl B YCl ₃ C YCl ₂ D Y ₂ Cl ₃	B	1
9	The strong bonds in the following substances: CO ₂ (g), HCl(g), CaO(s), Na(s) are, in order: A covalent, covalent, ionic, ionic B covalent, ionic, ionic, metallic C covalent, covalent, ionic, metallic D covalent, ionic, ionic, ionic.	C	1
10	Which of the following statements about selenium is NOT true? A Selenium is in period 3, group 16 in the Periodic Table. B When selenium atoms react with a metal, each selenium atom gains two electrons. C When selenium reacts with chlorine, it forms a molecular compound with the formula of SeCl ₂ . D Selenium would be expected to have a reasonably low melting point and to be a non-conductor of electricity.	A	1

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Section B—Written answers (37 marks)

11	Name the following compounds: a $\text{Fe}(\text{HCO}_3)_2$ b $(\text{NH}_4)_3\text{P}$ c N_2Cl_6	a Iron(II) hydrogencarbonate b Ammonium phosphide c Dinitrogen hexachloride	3
12	Give the formulas of the following compounds: a magnesium phosphate b diphosphorus trisulfide c ammonia d copper(I) sulfate.	a $\text{Mg}_3(\text{PO}_4)_2$ b P_2S_3 c NH_3 d Cu_2SO_4	4
13	a If the formula of magnesium thiocyanate is $\text{Mg}(\text{SCN})_2$, what is the formula of the thiocyanate ion? b i What is the formula of sodium sulfite? ii Using your answer to i, predict the formula of the tellurite ion?	a SCN^- b i Na_2SO_3 ii TeO_3^{2-}	3
14	Atoms W, X, Y and Z have the following electron configurations: W 2, 8, 2 X 2, 5 Y 2, 8 Z 2, 8, 7 a Which element, W, X, Y or Z, would be least reactive? b Which element is in group 15 in the periodic table? c Which could form a positive ion that has the same electron configuration as neon? d Give the formula of the substance formed when W and X react. e What type of substance would form when X reacts with Z?	a Y b X c W d W_3X_2 e Covalent molecular	5

Unit 2A, Atomic Structure and Bonding

To answer questions 15, 16, 17 and 18 refer to the information given in the table below:

Substance	Melting point ($^{\circ}\text{C}$)	Boiling point ($^{\circ}\text{C}$)	Solubility in water at 20°C (g substance per 100 g of water)	Solubility in water at 80°C (g substance per 100 g of water)
potassium bromide	734	1435	60	90
benzoic acid, $\text{C}_7\text{H}_6\text{O}_2$	122	249	0.29	4.0
methanol, CH_3OH	-98	65	very soluble	very soluble

15	a Describe a method you could use to obtain some pure benzoic acid from a mixture of potassium bromide and benzoic acid. b Explain, in terms of the compounds' properties why your method would be successful.	
	a Add the mixture to 100 mL of water at 80°C until no more will dissolve. Filter the hot mixture. Allow the filtrate to cool to around 20°C. The solid that separates from the solution will be benzoic acid. b Potassium bromide is very soluble in both hot and cold water. However, benzoic acid is much less soluble in cold water than in hot water. So, when the solution is cooled, the less soluble compound, the benzoic acid, will form crystals.	
16	Draw electron dot diagrams for potassium bromide and methanol.	
	$\left[\text{K} \right]^+ \left[: \ddot{\text{Br}} : \right]^-$ $\begin{array}{c} \text{H} \\ \vdots \\ \text{H} : \text{C} : \ddot{\text{O}} : \text{H} \\ \vdots \\ \text{H} \end{array}$	

Unit 2A, Atomic Structure and Bonding

17	Explain why potassium bromide has a much higher melting point than methanol.							
	Potassium bromide is an ionic compound with strong bonding throughout the three-dimensional crystal lattice. As a result of these strong bonds it takes a considerable amount of energy to break the lattice into its individual ions. Hence, the melting point of potassium bromide is high. Methanol is a covalent molecular compound. The covalent bonds between the individual atoms in the molecules are strong, but the intermolecular bonds (bonds between methanol molecules) are much weaker. When methanol melts it is these weak intermolecular bonds that are broken, and so the melting point is lower than potassium bromide.	4						
18	Compare the electrical conductivity of potassium bromide and methanol in solid and liquid states and in aqueous solution. Explain the differences and similarities in terms of the bonding and structure of the compounds. Neither solid potassium bromide nor solid methanol is a conductor of electricity. There are no free charge carriers (ions or electrons) present in the solid state of either compound. Methanol is not a conductor in the liquid state or in aqueous solution because, once again, there are no free charge carriers present (there are only neutral molecules). However, the liquid (molten) state of potassium bromide and the aqueous solution both conduct because of the presence of free ions that are able to move under the influence of an electric field.	3						
19	Some of the uses of graphite, an allotrope of carbon, and carbon dioxide, a compound of carbon, are given in the table below: <table><thead><tr><th></th><th>Uses</th></tr></thead><tbody><tr><td>Graphite</td><td> – lubricant – black colouring in ‘lead’ pencils – for making electrodes in cells, such as in the dry cell – used in the manufacture of crucibles in steel, brass and bronze industries. </td></tr><tr><td>Carbon dioxide</td><td> – as dry ice, a refrigerant used when refrigeration equipment is not practical – in fire extinguishers – in carbonated drinks – as a leavening (rising) agent in baking. </td></tr></tbody></table> Compare and explain these uses in terms of the properties, bonding and structure of the two carbon substances. Graphite: The lubricating property results from the layers of carbon atoms present in graphite having only weak attractive forces between them. The layer structure allows graphite to be used in ‘lead’ pencils where layers of graphite are left on the paper as the pencil is drawn across the surface. Graphite has delocalised electrons between the layers of covalently bonded carbon atoms, that allow it to be used as a cheap electrical conductor. The strong bonding between the carbon atoms in the covalent layer lattice ensures that graphite has a very high melting and boiling point. As a result it can be used to make equipment that is able to withstand high temperatures.		Uses	Graphite	– lubricant – black colouring in ‘lead’ pencils – for making electrodes in cells, such as in the dry cell – used in the manufacture of crucibles in steel, brass and bronze industries.	Carbon dioxide	– as dry ice, a refrigerant used when refrigeration equipment is not practical – in fire extinguishers – in carbonated drinks – as a leavening (rising) agent in baking.	8
	Uses							
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Unit 2A, Atomic Structure and Bonding

	Carbon dioxide: Solid carbon dioxide (dry ice) is a covalent molecular substance that sublimates at around -78°C. The low sublimation point is due to the weak intermolecular bonds. At this temperature it is an excellent refrigerant. More importantly, the fact that it sublimates means that there will be no liquid residue to clean up. Carbon dioxide does not support combustion and is denser than air. Because it is covalent molecular the bonds between molecules are weak and it is a gas at room temperature. The gas can be used to blanket a fire and prevent oxygen from reaching the fuel. The covalent molecular carbon dioxide is very slightly soluble in water and dissolves to produce a solution of a weak acid that gives carbonated drinks their unique 'taste'. Since carbon dioxide is a covalent molecular substance with a low melting and boiling point, it is produced as a gas in a chemical reaction, such as the baking of bread and cakes. This gas causes the bread or cake to rise.	
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Chemistry for WA 1 TEST 2

Unit 2A, Introduction to the Mole

Name: _____ Class: _____ Date: _____

Instructions: Write answers in the right-hand column.

Score: _____ / 25 marks

Section A—Multiple choice (5marks)

1	The relative atomic masses of carbon and magnesium are approximately 12 and 24 respectively. Which one of the following statements is true? A A carbon atom has half the mass of a magnesium atom. B An atom of carbon has a mass of 12 g and an atom of magnesium has a mass of 24 g. C Because the atomic mass ratio is 1:2, carbon and magnesium would combine to form the compound Mg_2C. D A magnesium atom has 12 more protons than a carbon atom.	A	1
2	Rubidium occurs in nature as a mixture of two isotopes with atomic masses of 85.02 and 87.01. What approximate percentage of each isotope must be present if the atomic mass of the element rubidium is 85.47? A 25% Rb-85, 75% Rb-87 B 50% Rb-85, 50% Rb-87 C 70% Rb-85, 30% Rb-87 D 95% Rb-85, 5% Rb-87	C	1
3	Four students were required to determine the calcium ion content in a sample of limestone. The answers they obtained are given below. Which of these answers could not possibly be correct? A 0.012 mol of calcium ions B 2.4×10^{-2} calcium ions C 0.47 g of calcium ions D 5.3×10^{21} calcium ions.	B	1
4	28 g of nitrogen gas (N_2): A contains 6.02×10^{23} atoms of nitrogen B contains 6.02×10^{23} molecules of nitrogen C contains two mole of nitrogen molecules D is heavier than 28 g of hydrogen.	B	1
5	Which of the following would have the largest mass? A 50.0 g of sodium chloride B 1.50 mol of sodium chloride C 6.02×10^{22} formula units of sodium chloride D the mass of sodium chloride that contains 1 mol of chloride ions.	B	1

Unit 2A, Introduction to the Mole

Section B—Written answers (20 marks)

6	a Calculate the amount, in mol, of aluminium ions in 100 g of aluminium sulfate. b What is the mass of 9.23×10^{22} formula units of lead(II) permanganate?	a $n(\text{Al}_2(\text{SO}_4)_3) = \frac{m(\text{Al}_2(\text{SO}_4)_3)}{M(\text{Al}_2(\text{SO}_4)_3)}$ $M(\text{Al}_2(\text{SO}_4)_3) = 342.14 \text{ g mol}^{-1}$ $n(\text{Al}_2(\text{SO}_4)_3) = \frac{100}{342.14} = 0.2923 \text{ mol}$ $n(\text{Al}^{3+}) = 2 \times n(\text{Al}_2(\text{SO}_4)_3) = 0.585 \text{ mol}$ b $n(\text{Pb}(\text{MnO}_4)_2) = \frac{N(\text{Pb}(\text{MnO}_4)_2)}{N_A}$ $n(\text{Pb}(\text{MnO}_4)_2) = \frac{9.23 \times 10^{22}}{6.022 \times 10^{23}}$ $= 0.1533 \text{ mol}$ $\text{mass}(\text{Pb}(\text{MnO}_4)_2) = n \times M(\text{Pb}(\text{MnO}_4)_2)$ $M(\text{Pb}(\text{MnO}_4)_2) = 445.08 \text{ g mol}^{-1}$ $\text{mass}(\text{Pb}(\text{MnO}_4)_2) = 0.1533 \times 445.08$ $= 68.2 \text{ g}$	6
7	A student places 10.0 g of iodine, I_2, in a beaker and then gently heats the beaker. 0.00970 mole of the iodine sublimes into the atmosphere during the heating. How many molecules of iodine, I_2, remain in the beaker after heating?	$n(\text{I}_2) \text{ present at start} = \frac{m(\text{I}_2)}{M(\text{I}_2)}$ $= \frac{10.0}{253.8} = 0.03940 \text{ mol}$ $n(\text{I}_2) \text{ remaining} = 0.03940 - 0.00970$ $= 0.02970 \text{ mol}$ $\text{number of molecules remaining} = n \times N_A$ $= 0.02970 \times 6.022 \times 10^{23}$ $= 1.79 \times 10^{22}$	3

Unit 2A, Introduction to the Mole

8	One atom of element X has a mass of 4.48×10^{-23} g. a What is the molar mass of the element X? b Identify element X.	a $n(X) = \frac{N(X)}{6.022 \times 10^{23}}$ $= \frac{1}{6.022 \times 10^{23}}$ $= 1.661 \times 10^{-24} \text{ mol}$ $n(X) = \frac{m(X)}{M(X)}$ $\text{molar mass of X} = \frac{4.48 \times 10^{-23}}{1.661 \times 10^{-24}}$ $= 27.0 \text{ g mol}^{-1}$ b Element X is aluminium,	3
9	18.9 g of NaCl and 0.256 mole of MgCl_2 are dissolved in water in a beaker. How many chloride ions are present in the water in the beaker?	$n(\text{NaCl}) = \frac{m(\text{NaCl})}{M(\text{NaCl})} = \frac{18.9}{58.44} = 0.3234 \text{ mol}$ $n(\text{Cl}^-) \text{ from NaCl} = 0.3234 \text{ mol}$ $n(\text{Cl}^-) \text{ from MgCl}_2 = 2 \times 0.256 = 0.512 \text{ mol}$ $\text{Total } n(\text{Cl}^-) = 0.8354 \text{ mol}$ $\text{number of chloride ions} = n(\text{Cl}^-) \times N_A$ $= 0.8354 \times 6.022 \times 10^{23}$ $= 5.03 \times 10^{23} \text{ ions}$	4

Unit 2A, Introduction to the Mole

10	Which of the compounds, NH_4NO_3 or Ca_3N_2 , has the larger percentage, by mass, of nitrogen?	NH_4NO_3 Percentage by mass = $\frac{\text{mass of nitrogen in 1 mole}}{M(\text{NH}_4\text{NO}_3)} \times 100$ $= \frac{28.02}{80.052} \times 100 = 35.0\%$ Ca_3N_2 Percentage by mass = $\frac{\text{mass of nitrogen in 1 mole}}{M(\text{Ca}_3\text{N}_2)} \times 100$ $= \frac{28.02}{148.26} \times 100 = 18.9\%$ Therefore, NH_4NO_3 has the highest percentage by mass of nitrogen.	4
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Chemistry for WA 1 TEST 3

Unit 2A, Gases, Liquids, Solids and Solutions

Name: _____ Class: _____ Date: _____

Instructions: Write answers in the right-hand column.

Score: _____ / 47 marks

Section A—Multiple choice (10 marks)

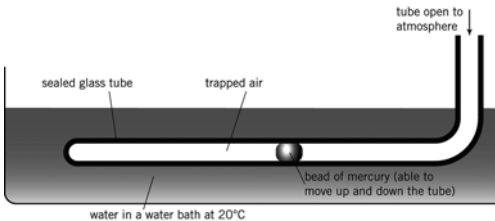
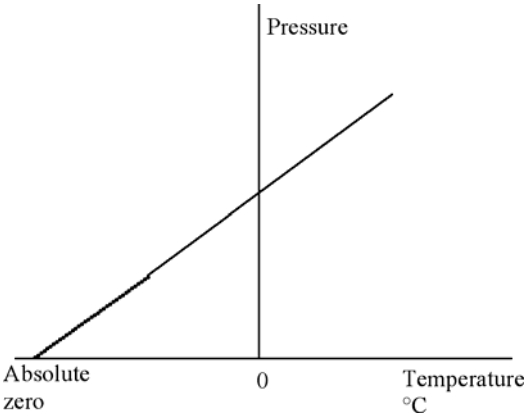
1	Which one of the following statements is INCORRECT? A Liquids and gases are both fluids and they can be poured. B Solids and liquids have a definite volume but gases have a variable volume. C Solids are incompressible but liquids and gases are highly compressible. D Solids have a definite shape, but liquids and gases have variable shapes.	C	1
2	A gas is confined in a cylinder with a movable piston at one end. Assuming the temperature remains constant, if the piston is pushed in, which of the following statements about the volume and pressure of the gas is correct? A The pressure will remain unchanged. B The pressure will increase because the volume is increasing. C The pressure will decrease due to the increase in volume. D Because of the decrease in volume, the pressure will increase.	D	1
3	A flask contains equal masses of two gases, hydrogen and nitrogen, at a temperature of 30°C. Which of the following statements about these gases is NOT correct? A The average kinetic energy ($\frac{1}{2}mv^2$) of the molecules of both gases is the same. B The average velocity of the H ₂ molecules is greater than the average velocity of the N ₂ molecules. C There are more H ₂ molecules than N ₂ molecules in the flask. D The pressure exerted by the H ₂ molecules in the flask is less than the pressure exerted by the N ₂ molecules.	D	1
4	Increasing the volume of a given amount of gas at a constant temperature causes the pressure to decrease because: A the molecules are striking a larger area with the same force. B there are fewer molecules. C the molecules are moving more slowly. D the molecules are hitting the wall with a smaller force.	A	1
5	Which of the following liquids has the highest vapour pressure? A pure water at 20°C B sea water at 40°C C pure water at 40°C D ethanol (boiling point 78°C) at 40°C.	D	1

Unit 2A, Gases, Liquids, Solids and Solutions

6	Which one of the following statements is true? The temperature at which ethanol boils: A gradually increases as the ethanol boils away B depends on room temperature C decreases if the atmospheric pressure falls D decreases if a small amount of water is added to it.	C	1
7	Which of the following statements about solutions are true? i When sodium chloride is dissolved in water, the water is the solute. ii An unsaturated solution contains more solute than is needed to saturate it. iii A supersaturated solution contains less dissolved solid per unit volume than a saturated solution. iv In general, the solubility of a gas in a liquid decreases with increasing temperature. v When sodium nitrate dissolves in water, dissociation occurs. A ii, iii and iv B i, ii and iv C ii and v D iv and v.	D	1
8	Which of the following pairs of solutions would not form a precipitate when mixed together? A magnesium chloride solution and potassium sulfide solution B zinc chloride solution and potassium phosphate solution C calcium chloride solution and copper(II) nitrate solution D lead nitrate solution and iron(III) iodide solution.	C	1
9	Barium nitrate and barium chloride were mixed in water to form a solution X. Sodium carbonate and potassium chloride were mixed in water to form a solution Y. When X and Y were mixed, a white precipitate was formed. What is this white precipitate? A barium sulfate B barium carbonate C sodium chloride D potassium carbonate.	B	1
10	Of the following solutions, which would be best classified as a concentrated solution of a weak electrolyte: A 0.10 mol L ⁻¹ nitric acid B 6.0 mol L ⁻¹ potassium hydroxide C 5.0 mol L ⁻¹ acetic acid D 1.0 mol L ⁻¹ sugar.	C	1

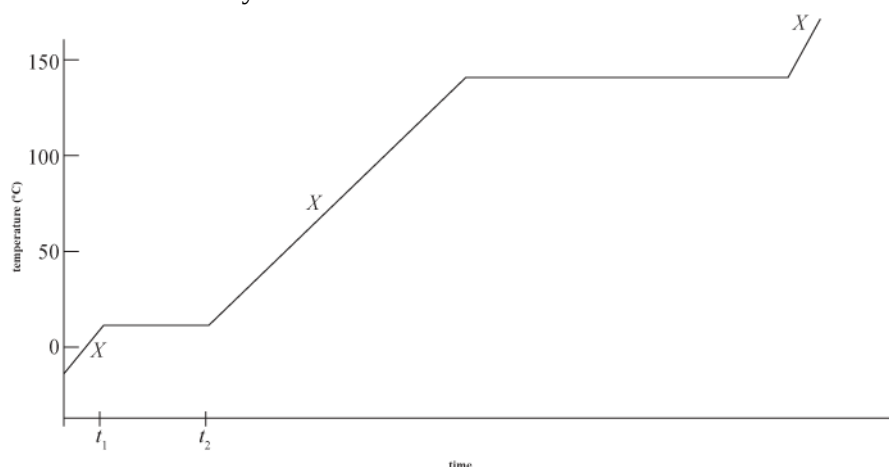
Unit 2A, Gases, Liquids, Solids and Solutions

Section B—Written answers (37 marks)

11	a Use the assumptions of the kinetic theory of gases to explain why raising the temperature of a constant mass of gas, at constant pressure, will result in an increase in the volume occupied by the gas. b Sketch an equipment set-up that could be used to demonstrate the property of gases described in part a. Label your diagram clearly. c Describe how the pressure would be kept constant in the equipment set-up you have sketched in part b.	a When the temperature of a sample of gas is increased, the average kinetic energy of the particles in the gas is increased. This means that the particles are exerting a larger force on the surface of the container in which they are held. If the pressure is to remain constant, the larger force needs to be spread over a larger surface area. In other words, the volume of the container must increase. b  c In the apparatus drawn, the pressure is kept constant by allowing the mercury in the tube to move as the temperature of the gas trapped in the tube is increased. When the mercury stops moving the pressure of the gas in the tube is the same as atmospheric pressure. Provided that atmospheric pressure does not change during the experiment, the apparatus allows the investigation of change in volume with temperatures at constant pressure.	6
12	Sketch a graph that shows the relationship between pressure (vertical axis) and temperature, in degrees Celsius, (horizontal axis) for a particular amount of gas at a constant volume.		2

Unit 2A, Gases, Liquids, Solids and Solutions

- 13** This heating curve shows the changes in temperature that occurred during the heating of a solid, as it melted and finally boiled.



- | | |
|---|---|
| <p>a What is the approximate boiling point of the substance?</p> <p>b Indicate on the graph one time when the kinetic energy of the substance is changing but its potential energy is remaining approximately constant.</p> <p>c During times t_1 to t_2, despite the substance being heated, its temperature is remaining constant. Explain this observation.</p> <p>d Would you expect the substance being heated to be tribromomethane (CHBr_3) or sodium bromide? Explain your reasoning.</p> | <p>a 140°C</p> <p>b Marked with an X (any one)</p> <p>c As the solid is melting, the heat energy is being used to disrupt the bonds between the particles (molecules) in the solid rather than to change their kinetic energy. Since the kinetic energy is not changing (but the potential energy is), the temperature of the substance does not change.</p> <p>d Tribromomethane – because this is a covalent molecular substance which is expected to have relatively low melting and boiling points. Sodium bromide is ionic and its melting and boiling points would be much higher than shown on the graph. Much more energy is required to break the strong ionic bonds in the lattice of positive and negative ions in sodium bromide, compared to the energy required to break the weak bonds between the neutral tribromomethane molecules.</p> |
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Unit 2A, Gases, Liquids, Solids and Solutions

14	Using the kinetic theory of matter, explain why washing hanging on a clothes line will dry quicker on a hot day than on a cold day.	The rate of evaporation of liquid depends on the temperature. Since temperature is a measure of the kinetic energy of the particles, at a higher temperature there will be more molecules in the liquid with enough kinetic energy to break away from the liquid and enter the gas phase.	2
15	During operation, the pressure in the radiator of a car is considerably higher than atmospheric pressure. Explain why it is dangerous to remove the radiator cap when the engine is still hot.	The increased pressure in the radiator of the car allows the water to be heated to a temperature greater than 100°C without boiling. If the radiator cap is removed while the temperature of the water is higher than 100°C the rapid decrease in pressure from that in the radiator to atmospheric pressure will cause the water to boil violently. The resulting steam or water that will 'burst' from the radiator, at temperatures in excess of 100°C, can cause severe burns.	2
16	When gaseous HBr is added to water, the solution formed is a very good conductor of electricity. However, when liquid CH ₃ OH is added to water, the solution formed does not conduct electricity. Explain these observations.	When HBr is added to water it reacts with the water and undergoes complete ionisation (it is a strong electrolyte). $\text{HBr(g)} + \text{H}_2\text{O(l)} \rightarrow \text{H}_3\text{O}^+(\text{aq}) + \text{Br}^-(\text{aq})$ (or $\text{HBr(g)} + \text{aq} \rightarrow \text{H}^+(\text{aq}) + \text{Br}^-(\text{aq})$) The presence of ions in solution makes it a good conductor. When methanol dissolves in water there is no reaction because methanol is a non-electrolyte. The methanol molecules simply mix with the water molecules and no ions are formed.	4

Unit 2A, Gases, Liquids, Solids and Solutions

17	Describe what you would expect to see when the following pairs of solutions are mixed: a a solution of cobalt nitrate and a solution of sodium carbonate b a solution of zinc chloride and a solution of calcium nitrate c a solution of lead nitrate and a solution of potassium sulfate d a solution of iron(III) chloride and a solution of sodium hydroxide.	a A pink solution is added to a colourless solution and a pink solid forms. b Two colourless solutions are mixed and there is no noticeable change. c Two colourless solutions are mixed and a white solid is formed. d A colourless solution is added to a brown solution and a brown solid is formed.	4
18	What mass of K_2SO_4 would be contained in 208 mL of $0.910 \text{ mol L}^{-1} K_2SO_4$ solution?	$n(K_2SO_4) = cV$ $= 0.910 \times 0.208$ $= 0.1893 \text{ mol}$ $m(K_2SO_4) = n \times M$ $= 0.1893 \times 174.26$ $= 33.0 \text{ g}$	2

Unit 2A, Gases, Liquids, Solids and Solutions

19	0.200 moles of solid MgCl_2 are added to a 1.20 L of 1.40 mol L^{-1} NaCl solution. a Calculate the concentration, in mol L^{-1}, of Cl^- ions in the 'new' solution. b What assumptions have been made in this calculation?	a $n(\text{Cl}^-)$ from $\text{MgCl}_2 = 0.400 \text{ mol}$ $n(\text{Cl}^-)$ from NaCl = cV $= 1.40 \times 1.20$ $= 1.680 \text{ mol}$ total amount of $\text{Cl}^- = 2.08 \text{ mol}$ in a volume of 1.20 L concentration of $\text{Cl}^- = \frac{n(\text{Cl}^-)}{V}$ $= \frac{2.08}{1.20}$ $= 1.73 \text{ mol L}^{-1}$ b Assumptions include: – the volume of solution does not change – all the magnesium chloride dissolved – there is no other reaction between magnesium chloride and sodium chloride.	5
20	The health authorities in Australia recommend the safe level of lead in drinking water to be 0.01 ppm (1 ppm of lead = 1 mg of lead dissolved in 1 kg of drinking water). Despite lead and its compounds being very toxic, products used every day, such as car batteries, still contain lead. Lead and many of its compounds are insoluble in water, so it is assumed that very little lead enters the drinking water supply by the dissolution of lead substances. Lead(II) oxide, a compound regarded as insoluble in water, is used to make crystal glassware and as a pottery glaze. The solubility of lead oxide at 25°C is 0.002 g per 100 g of water. If some water is stored in a bowl of crystal glass, determine whether this water could possibly dissolve sufficient lead oxide to raise the level of lead in the water over the recommended safe level.	0.002 g of lead oxide per 100 g of water is equivalent to 0.02 g of lead oxide in 1000 g (1 kg) of water. $n(\text{PbO}) = \frac{m(\text{PbO})}{M(\text{PbO})}$ $= \frac{0.02}{223.2}$ $= 8.96 \times 10^{-5}$ $= \text{mol of Pb in 0.02 g of lead oxide}$ $m(\text{Pb}) = 8.96 \times 10^{-5} \times 207.2$ $= 0.019 \text{ g} = 19 \text{ mg}$ Hence, there is 19 mg of lead in 1 kg of water which is 19 ppm of lead. This is above the recommended safe level.	4

Chemistry for WA 1 TEST 4

Unit 2A, Chemical Reactions

Name: _____ Class: _____ Date: _____

Instructions: Write answers in the right-hand column.

Score: _____ / 32 marks

Section A—Multiple choice (5 marks)

1	The following equation represents the decomposition of lead(II) nitrate on heating $x \text{Pb(NO}_3)_2 \rightarrow y \text{PbO} + z \text{NO}_2 + \text{O}_2$ The respective values of x, y and z are: A 1, 2, 2 B 2, 2, 4 C 4, 4, 8 D 2, 3, 4	B	1
2	Which of these is an <u>incorrect</u> interpretation of this balanced equation? $2\text{S(s)} + 3\text{O}_2\text{(g)} \rightarrow 2\text{SO}_3\text{(g)}$ A 2 atoms S + 3 molecules $\text{O}_2 \rightarrow$ 2 molecules SO_3 B 2 g S + 3 g $\text{O}_2 \rightarrow$ 2 g SO_3 C 2 mol S + 3 mol $\text{O}_2 \rightarrow$ 2 mol SO_3 D none of the above.	B	1
3	54 g of hydrogen cyanide, HCN, combined with 80 g of oxygen to produce 18 g of water, 28 g of nitrogen and carbon dioxide, but nothing else. The carbon dioxide had a mass of: A 44 g B 88 g C 134 g D 286 g	B	1
4	When one mole of a particular compound containing only carbon and hydrogen is burnt in excess pure oxygen, equal amounts, in mole, of CO_2 and H_2O are formed. The hydrocarbon could be: A CH_4 B C_2H_6 C C_2H_4 D C_6H_6	C	1
5	A solution of sodium hydroxide when added to a solution of aluminium chloride produces a gel-like precipitate. The precipitate reaction could be represented as: A $\text{Na}^+(\text{aq}) + \text{Cl}^-(\text{aq}) \rightarrow \text{NaCl(s)}$ B $\text{Al}^{3+}(\text{aq}) + 3\text{OH}^-(\text{aq}) \rightarrow \text{Al(OH)}_3\text{(s)}$ C $\text{Al}^{3+}(\text{aq}) + 3\text{Cl}^-(\text{aq}) \rightarrow \text{AlCl}_3\text{(s)}$ D $\text{H}^+(\text{aq}) + \text{OH}^-(\text{aq}) \rightarrow \text{H}_2\text{O(l)}$	B	1

Unit 2A, Chemical Reactions

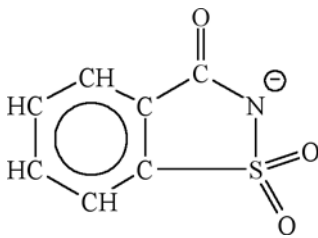
Section B—Written answers (27 marks)

6	Write balanced molecular equations for each of the following reactions: a aluminium is heated with steam to give hydrogen gas and solid aluminium oxide b chlorine gas is bubbled through a solution of sodium iodide to form solid iodine and a solution of sodium chloride c solid manganese(IV) oxide (MnO_2) is added to dilute hydrochloric acid to produce water, chlorine gas and a solution of manganese(II) chloride d solutions of calcium hydroxide and phosphoric acid are mixed to give a precipitate of calcium phosphate, and water e copper(II) oxide reacts with ammonia gas to produce copper, nitrogen gas and water.	a $2\text{Al(s)} + 3\text{H}_2\text{O(g)} \rightarrow \text{Al}_2\text{O}_3\text{(s)} + 3\text{H}_2\text{(g)}$ b $\text{Cl}_2\text{(g)} + 2\text{NaI(aq)} \rightarrow \text{I}_2\text{(s)} + 2\text{NaCl(aq)}$ c $\text{MnO}_2\text{(s)} + 4\text{HCl(aq)} \rightarrow \text{Cl}_2\text{(g)} + 2\text{H}_2\text{O(l)} + \text{MnCl}_2\text{(aq)}$ d $3\text{Ca(OH)}_2\text{(aq)} + 2\text{H}_3\text{PO}_4\text{(aq)} \rightarrow \text{Ca}_3\text{(PO}_4)_2\text{(s)} + 6\text{H}_2\text{O(l)}$ e $3\text{CuO(s)} + 2\text{NH}_3\text{(g)} \rightarrow 3\text{Cu(s)} + \text{N}_2\text{(g)} + 3\text{H}_2\text{O(l)}$	5
7	When butane is burnt in air, the following reaction occurs: $2\text{C}_4\text{H}_{10}\text{(g)} + 13\text{O}_2\text{(g)} \rightarrow 8\text{CO}_2\text{(g)} + 10\text{H}_2\text{O(g)}$ What mass of carbon dioxide would be formed from the burning of 143 g of butane in excess oxygen?	$n(\text{butane}) = \frac{m(\text{C}_4\text{H}_{10})}{M(\text{C}_4\text{H}_{10})}$ $= \frac{143}{58.12} = 2.460 \text{ mol}$ From equation, 2 mol C_4H_{10} produces 8 mol CO_2, so 1 mol C_4H_{10} produces 4 mol CO_2. $n(\text{CO}_2) \text{ formed} = 4 \times 2.460$ $= 9.840 \text{ mol}$ $\text{mass}(\text{CO}_2) \text{ formed} = n \times M$ $= 9.840 \times 44.01$ $= 433 \text{ g}$	4

Unit 2A, Chemical Reactions

8	What is the concentration, in mol L⁻¹, of a K₂C₂O₄ solution if 35.00 mL of it is required to react completely with 47.65 mL of 0.0632 mol L⁻¹ KMnO₄ solution? $2\text{MnO}_4^-(\text{aq}) + 5\text{C}_2\text{O}_4^{2-}(\text{aq}) + 16\text{H}^+(\text{aq}) \rightarrow 2\text{Mn}^{2+}(\text{aq}) + 10\text{CO}_2(\text{g}) + 8\text{H}_2\text{O}(\text{l})$		
	$n(\text{KMnO}_4) = cV = 0.0632 \times 0.04765$ $= 0.003011 \text{ mol} = \text{mol of MnO}_4^-$ From equation, 2 mol of MnO₄⁻ reacts completely with 5 mol of C₂O₄²⁻ so 1 mol of MnO₄⁻ reacts completely with $\frac{5}{2}$ mol of C₂O₄²⁻ therefore 0.003011 mol of MnO₄⁻ reacts completely with $\frac{5}{2} \times 0.003011$ $= 0.007528 \text{ mol of C}_2\text{O}_4^{2-}$ That is, mol of K₂C₂O₄ required = 0.007528 $c(\text{K}_2\text{C}_2\text{O}_4) = \frac{n(\text{K}_2\text{C}_2\text{O}_4)}{V} = \frac{0.007528}{0.03500}$ $= 0.215 \text{ mol L}^{-1}$	4	
9	Evil smelling hydrogen sulfide (bad-egg gas), H₂S, can be destroyed by reacting it with oxygen according to the equation: $2\text{H}_2\text{S}(\text{g}) + 3\text{O}_2(\text{g}) \rightarrow 2\text{SO}_2(\text{g}) + 2\text{H}_2\text{O}(\text{l})$ What mass of oxygen will be left over when 16.0 g of hydrogen sulfide reacts with 30.0 g of oxygen?	$n\text{H}_2\text{S} = \frac{m(\text{H}_2\text{S})}{M(\text{H}_2\text{S})} = \frac{16.0}{34.076} = 0.4695 \text{ mol}$ From equation, 2 mol H₂S combines with 3 mol O₂ so 1 mol H₂S combines with $\frac{3}{2}$ mol O₂ 0.4695 mol of H₂S reacts with 0.7042 mol O₂ $m(\text{O}_2) = n(\text{O}_2) \times M(\text{O}_2)$ $= 0.7042 \times 32.00 = 22.53 \text{ g}$ Mass of O₂ remaining is 30.0 - 22.53 $= 7.47 \text{ g}$	4

Unit 2A, Chemical Reactions

10	What is the concentration, in g L^{-1}, of a solution of hydrochloric acid if 30.0 mL of this solution reacted completely with 40.0 mL of $0.244 \text{ mol L}^{-1} \text{ NaOH}$? $\text{H}^+(\text{aq}) + \text{OH}^-(\text{aq}) \rightarrow \text{H}_2\text{O}$	$n(\text{NaOH}) = cV = 0.244 \times 0.0400$ $= 0.009760 \text{ mol}$ $= n(\text{OH}^-)$ From equation, $n(\text{H}^+) = n(\text{OH}^-)$ $= 0.009760$ $= n(\text{HCl})$ $n(\text{HCl}) = \frac{m(\text{HCl})}{M(\text{HCl})}$ $m(\text{HCl}) = 0.009760 \times 36.458$ $= 0.3558 \text{ g}$ $c(\text{HCl}) = \frac{m(\text{HCl})}{V}$ $= \frac{0.3558}{0.0300} = 11.9 \text{ g L}^{-1}$	4
11	A liquid sweetener contains the artificial sweetener calcium saccharin. The negative ion of saccharin is shown below:  To determine the percentage, by mass, of calcium saccharin in the liquid sweetener, 52.4 g of the liquid sweetener was reacted with a strong oxidising agent to convert all the sulfur present in the saccharin ions to sulfate ions. A solution of barium chloride was then added to form an insoluble solid called barium sulfate, from all the sulfate ions. The insoluble barium sulfate solid was filtered from the mixture, dried and weighed. Its mass was 1.27 g: a What is the formula of calcium saccharin? b How many mole of barium sulfate was formed from the liquid sweetener? c How many mole of sulfur atoms are contained in this amount of barium sulfate?	a $\text{Ca}(\text{C}_7\text{H}_4\text{NO}_3\text{S})_2$ b $n(\text{BaSO}_4) = \frac{\text{mass}(\text{BaSO}_4)}{M(\text{BaSO}_4)} = \frac{1.27}{233.36}$ $= 0.00544 \text{ mol}$ c $n(\text{S}) = n(\text{BaSO}_4) = 0.00544 \text{ mol}$ d 2 mol of S is contained in 1 mol of calcium saccharin or 1 mol of S is contained in $\frac{1}{2}$ mol of calcium saccharin $n(\text{calcium saccharin}) = \frac{1}{2} \times n(\text{S})$ $= 0.00272 \text{ mol}$ $m(\text{calcium saccharin}) = n \times M$ $= 0.00272 \times 404.424$ $= 1.10 \text{ g}$ e Percentage of calcium saccharin $= \frac{\text{mass of calcium saccharin}}{\text{total mass}} \times 100$ $= \frac{1.10}{52.4} \times 100 = 2.10\%$	6

Unit 2A, Chemical Reactions

	d Assuming all the sulfur atoms in the barium sulfate were originally present in the calcium saccharin, calculate the mass of calcium saccharin in the 52.4 g sample of the liquid sweetener. e What is the percentage, by mass, of calcium saccharin in the liquid sweetener?		
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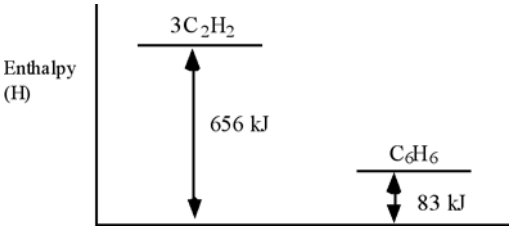
Unit 2A, Reactions – Energy and Rates

Name: _____ Class: _____ Date: _____

Instructions: Write answers in the right-hand column.

Score: _____ / 23 marks

Section A—Multiple choice (10 marks)

1	If the following reactions were carried out in thermally insulated containers, for which ones would the temperature of the mixture increase as the reaction occurs? i $\text{Zn(s)} + \text{Br}_2(\text{l}) \rightarrow \text{ZnBr}_2(\text{s}); \Delta H = -329 \text{ kJ}$ ii $\text{Na}_2\text{S}_2\text{O}_3 \cdot 5\text{H}_2\text{O(s)} + 45 \text{ kJ} \rightarrow \text{Na}_2\text{S}_2\text{O}_3(\text{aq}) + 5\text{H}_2\text{O(l)}$ iii $\text{PBr}_5(\text{g}) \rightarrow \text{PBr}_3(\text{g}) + \text{Br}_2(\text{g}); \Delta H = +105 \text{ kJ}$ iv $\text{Pb}^{2+}(\text{aq}) + 2\text{I}^-(\text{aq}) \rightarrow \text{PbI}_2(\text{s}) + 63 \text{ kJ}$ A ii and iii B i and ii C iii and iv D i and iv	D	1
2	 Use the information in the above diagram to calculate ΔH for the reaction $3\text{C}_2\text{H}_2(\text{g}) \rightarrow \text{C}_6\text{H}_6(\text{g})$ A +739 kJ B +573 kJ C –573 kJ D –656 kJ	C	1
3	Magnesium combines with oxygen according to the equation $2\text{Mg(s)} + \text{O}_2(\text{g}) \rightarrow 2\text{MgO(s)} + 1204 \text{ kJ}$ Which one of the following statements concerning this reaction is true? A The reaction is endothermic. B The temperature of the surroundings will decrease when the reaction occurs. C The enthalpy of the product will be less than the enthalpy of the reactants. D The energy required to break the bonds in magnesium and oxygen will be greater than the energy released in the forming of the bonds in magnesium oxide.	C	1

Unit 2A, Reactions – Energy and Rates

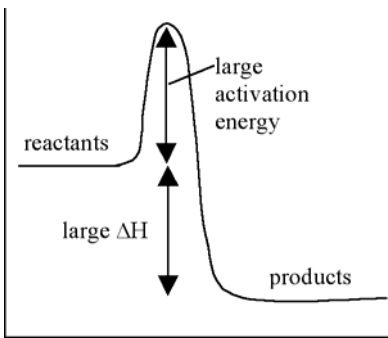
	Questions 4 and 5 refer to the following graph which shows the volume of hydrogen produced against time during a rate of reaction experiment between zinc and dilute hydrochloric acid.		
4	At which point in the graph is the reaction rate the greatest?	A	1
5	At which point is the reaction complete?	D	1
6	In a chemical reaction it is correct to say that (choose one answer): A the activated complex has a lower energy than either the reactants or products. B most collisions involve only two particles. C a successful collision is one in which the reactants are re-formed. D the rate can be determined by inspection of the stoichiometric equation.	B	1
7	‘The rates of all chemical reactions increase with a rise in temperature.’ This statement is (choose one answer): A true, since the activation energies for all reactions increase with a rise in temperature. B false, since the rate of an exothermic reaction decreases with a rise in temperature. C false, since the rate of an endothermic reaction decreases with a rise in temperature. D true, since a greater proportion of molecular collisions will result in reaction.	D	1
8	For the reaction $2\text{NH}_3(\text{g}) \rightarrow \text{N}_2(\text{g}) + 3\text{H}_2(\text{g})$; $\Delta H = +92 \text{ kJ}$, which one of the following statements is correct? A When 4 mole of ammonia decomposes, 92 kJ of heat will be evolved. B When 1 mole of nitrogen reacts with 3 moles of hydrogen, 92 kJ of heat will be absorbed by the system. C When 1 mole of ammonia reacts, 46 kJ of heat will be absorbed from the surroundings. D When 2 mole of ammonia decomposes, the temperature will decrease by 92°C.	C	1

Unit 2A, Reactions – Energy and Rates

	The following information refers to questions 9 and 10. Examine the following potential energy diagram for a chemical reaction:		
9	The enthalpy change ΔH and the activation energy E_a in kJ for the reaction are respectively A $\Delta H = -25 \text{ kJ}$ and $E_a = 35 \text{ kJ}$ B $\Delta H = +25 \text{ kJ}$ and $E_a = 35 \text{ kJ}$ C $\Delta H = 35 \text{ kJ}$ and $E_a = 60 \text{ kJ}$ D $\Delta H = -60 \text{ kJ}$ and $E_a = 35 \text{ kJ}$	A	1
10	If a catalyst is added to the system (choose one answer): A both ΔH and E_a decrease B ΔH remains constant but E_a decreases C ΔH decreases and E_a remains constant D both ΔH and E_a remain constant.	B	1

Unit 2A, Reactions – Energy and Rates

Section B—Written answers (13 marks)

11	A compound of antimony, antimony sulfide, is used to make safety matches. A small amount of antimony sulfide is mixed with potassium chlorate and glued to the tip of the matchstick. When the match is struck on the side of the matchbox, the red phosphorus on the side of the box ignites, which in turn ignites the antimony sulfide/potassium chlorate mixture. Heat from this reaction then makes the match wood burn. a Predict the formula of antimony sulfide (which contains the Sb^{3+} ion) and potassium chlorate. b When antimony sulfide and potassium chlorate react, they form antimony oxide, potassium chloride and sulfur dioxide. Write a balanced equation for this reaction. c Is this reaction an exothermic reaction or an endothermic reaction? Explain how you decided your answer. d Sketch an energy profile diagram, for the reaction of antimony sulfide with potassium chlorate, indicating the relative potential energies of the reactants and products and the activation energy. e Using the collision theory, explain why antimony sulfide and potassium chlorate do not react when the match is stored under normal conditions, but once struck on the side of the box, a reaction occurs.	a Sb_2S_3; KClO_3 b $\text{Sb}_2\text{S}_3(\text{s}) + 3\text{KClO}_3(\text{s}) \rightarrow 3\text{KCl}(\text{s}) + \text{Sb}_2\text{O}_3(\text{s}) + 3\text{SO}_2(\text{g})$ c Exothermic because when the antimony sulfide/ potassium chlorate mixture reacts, there is sufficient heat to make the wood in the match burn d  e The heat produced from the burning of the phosphorus (on the side of the box) increases the average kinetic energy of the antimony sulfide and potassium chlorate particles. As a result, more of these particles will have the required activation energy, when they collide, for a reaction to occur.	7
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Unit 2A, Reactions – Energy and Rates

12	The fizzy sensation from sherbet lolly powder is caused by a gas produced in a reaction between two substances. The sherbet powder consists of the two dry reactants citric acid, $C_6H_8O_7$, and sodium hydrogencarbonate, and a sweetener such as icing sugar. The gas-producing chemical reaction does not begin until the dry ingredients become wet. As soon as they dissolve into the saliva in your mouth, the reaction begins. In this reaction the citric acid reacts with the hydrogencarbonate ions to form carbon dioxide (the fizz), water and citrate ions, $C_6H_5O_7^{3-}$. a Explain in terms of the collision theory why a reaction occurs much quicker when the two reactants are ‘wet’. b If some sherbet powder were to be poured into hot water, would you expect the rate of ‘fizz production’ to change? Explain your answer in terms of the collision theory.	a The reactants, citric acid and sodium hydrogencarbonate must be close enough to collide and have enough energy to overcome the activation energy barrier. This is more likely to take place in solution where the dissolved citric acid molecules and hydrogencarbonate ions are moving around and are more likely to undergo collision. b The average kinetic energy of the particles is greater at the higher temperature. The reaction rate would be increased at the higher temperature as there will be more collisions between the faster moving particles and, more importantly, these particles will have more energy and so the chance of collisions with sufficient activation energy to react will be greater.	4
13	Acetic acid, CH_3COOH, is industrially manufactured from methanol according to the following reaction $CH_3OH(l) + CO(g) \rightarrow CH_3COOH(l);$ $\Delta H = -137 \text{ kJ mol}^{-1}$ The most modern chemical plants now use rhodium, Rh, as a catalyst for this reaction. a i What is meant by the term ‘$\Delta H = -137 \text{ kJ mol}^{-1}$’ written after the chemical equation. ii Calculate the amount of heat released or absorbed when 100 g of acetic acid is formed in the reaction given above.	a i When the reaction takes place according to the stoichiometry of the given equation, then 137 kJ of heat energy will be released to the surroundings. ii $n(\text{acetic acid}) = \frac{\text{mass}(CH_3COOH)}{M(CH_3COOH)}$ $= \frac{100}{60.052} = 1.665 \text{ mol}$ Since 137 kJ is produced when 1 mole of acetic acid is formed, the energy released when 1.665 mole of acetic acid is formed, $= 1.665 \times 137 = 228 \text{ kJ}$	2

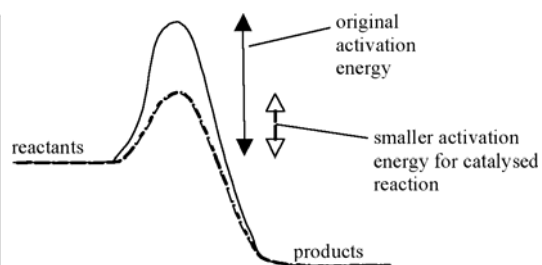
Unit 2A, Reactions – Energy and Rates

b

- i** What is the role of a catalyst in a reaction?
- ii** In terms of the collision theory, explain how a catalyst affects the rate of a reaction. Include an energy profile of the reaction in your answer.

b

- i** A catalyst provides an alternative reaction pathway with lower activation energy for reaction.
- ii** Since the catalyst provides a pathway with a lower activation energy it is more likely that collisions between reacting particles will have enough energy to provide this activation energy. Thus the rate of reaction will be greater.



Chemistry for WA 1 TEST 6

Unit 2B, Chemical Reactions Revisited

Name: _____ Class: _____ Date: _____

Instructions: Write answers in the right-hand column.

Score: _____ / 40 marks

Section A—Multiple choice (8 marks)

1	For which of the following reactions would the following observation apply: two colourless solutions are mixed to form a white precipitate. A a solution of magnesium nitrate is mixed with a solution of sodium sulfate B solutions of potassium carbonate and nickel chloride are mixed C dilute hydrochloric acid is added to a solution of silver nitrate D a solution of sodium hydroxide is added to dilute nitric acid	C	1
2	Mercury is obtained by roasting the natural ore cinnabar, which is red mercury sulfide. This process can be represented as: mercury sulfide + oxygen → mercury + sulfur dioxide 1 kg of cinnabar when roasted yields about 860 g of mercury. It follows that the mass of sulfur dioxide will be: A negligible because it is a gas B less than the mass of oxygen consumed C 140 g D more than 140 g.	D	1
3	The reaction of sodium sulfite with hydrochloric acid may be represented by the equation: $\text{Na}_2\text{SO}_3(\text{s}) + 2\text{H}^+(\text{aq}) \rightarrow 2\text{Na}^+(\text{aq}) + \text{H}_2\text{O}(\text{l}) + \text{SO}_2(\text{g}).$ On the basis of the information in the equation, which one of the following is true? A When 1 mol of Na_2SO_3 is treated with acid, 1 mole of SO_2 will always be produced. B Na_2SO_3 and H^+ must always be present in the mole ratio of 1:2 for a reaction to take place. C The ratio by mass of Na_2SO_3 to H^+ in a reaction mixture containing the correct stoichiometric proportions is 1:2. D 0.4 mole of H_2O will be produced when 0.8 mole of H^+ reacts completely with excess Na_2SO_3.	D	1
4	When solutions containing calcium ions and phosphate ions are mixed, the following precipitation reaction occurs: $3\text{Ca}^{2+}(\text{aq}) + 2\text{PO}_4^{3-}(\text{aq}) \rightarrow \text{Ca}_3(\text{PO}_4)_2(\text{s})$ How many mole of precipitate would form if solutions containing 0.825 mole of Ca^{2+} and 0.520 mole of PO_4^{3-} were mixed? A 0.260 mol B 0.275 mol C 0.520 mol D 0.825 mol	A	1

Unit 2B, Chemical Reactions Revisited

5	Hydrogen gas and iodine solid react to form hydrogen iodide according to the equation: $\text{H}_2(\text{g}) + \text{I}_2(\text{s}) \rightarrow 2\text{HI}(\text{g})$ 25.4 g of I_2 and 0.4 g of H_2 are reacted together and 25.6 g of HI forms. Assuming the reactant that is not in excess reacts completely, the mass of the excess reactant left at the end of the reaction is: A 0.2 g B 0.4 g C 0.8 g D 0.6 g	A	1
6	When 0.200 mol of calcium is mixed with 880 g of water, 4.48 L of hydrogen gas form (at STP). How would the volume of hydrogen produced change if the mass of water were to be decreased to 440 g? $\text{Ca}(\text{s}) + 2\text{H}_2\text{O}(\text{l}) \rightarrow \text{Ca}^{2+}(\text{aq}) + 2\text{OH}^{-}(\text{aq}) + \text{H}_2(\text{g})$ A Only one half the volume of hydrogen would be produced. B The volume of hydrogen produced would be the same. C The volume of hydrogen produced would double. D No hydrogen would be produced.	B	1
7	In an experiment to determine the empirical formula of a chloride of a metal X, a student reduced 0.277 g of the chloride of X to the metal X itself, by using aluminium powder as the reducing agent. She obtained 0.207 g of X. (relative atomic masses: $X = 207$; $\text{Cl} = 35$) From these results, the formula of the compound is shown to be: A XCl B XCl_2 C XCl_3 D XCl_4	B	1
8	0.305 g of a volatile hydrocarbon vaporises to 85.0 mL of vapour measured at room temperature and pressure. What is the molecular formula of the hydrocarbon? (molar volume at room temp and pressure = 24.0 L) A C_5H_{10} B C_5H_{12} C C_6H_{12} D C_6H_{14}	D	1

Unit 2B, Chemical Reactions Revisited

Section B—Written answers (32 marks)

9	Write balanced ionic equations for the following reactions: a solid sodium sulfite is added to dilute hydrochloric acid to produce water, sulfur dioxide gas and a solution of sodium chloride b solutions of iron(III) nitrate and potassium hydroxide are mixed c solutions of calcium hydroxide and phosphoric acid are mixed to give a precipitate of calcium phosphate, and water.	a $\text{Na}_2\text{SO}_3(\text{s}) + 2\text{H}^+(\text{aq}) \rightarrow \text{H}_2\text{O}(\text{l}) + \text{SO}_2(\text{g}) + 2\text{Na}^+(\text{aq})$ b $\text{Fe}^{3+}(\text{aq}) + 3\text{OH}^-(\text{aq}) \rightarrow \text{Fe}(\text{OH})_3(\text{s})$ c $3\text{Ca}^{2+}(\text{aq}) + 3\text{OH}^-(\text{aq}) + \text{H}_3\text{PO}_4(\text{aq}) \rightarrow \text{Ca}_3(\text{PO}_4)_2(\text{s}) + 3\text{H}_2\text{O}(\text{l})$	3
10	2.53 g of anhydrous sodium carbonate is made up to 500 mL of aqueous solution. What volume of 0.117 mol L^{-1} hydrochloric acid solution will be necessary to react completely with 25.0 mL of the sodium carbonate solution? $2\text{H}^+(\text{aq}) + \text{CO}_3^{2-}(\text{aq}) \rightarrow \text{CO}_2(\text{g}) + \text{H}_2\text{O}(\text{l})$	$M(\text{Na}_2\text{CO}_3)$ $= (2 \times 22.99) + 12.01 + (3 \times 16.00)$ $= 105.99 \text{ g mol}^{-1}$ $n(\text{Na}_2\text{CO}_3) \text{ in } 500 \text{ mL} = \frac{2.53}{105.99}$ $= 0.02387 \text{ mol}$ $n(\text{Na}_2\text{CO}_3) \text{ in } 25.0 \text{ mL} = 0.02387 \times \frac{25.0}{500}$ $= 0.001194$ $= \text{mol of } \text{CO}_3^{2-}$ $n(\text{H}^+) \text{ required} = 2 \times 0.001194$ $= 0.002388$ $= n(\text{HCl})$ $V(\text{HCl}) = \frac{0.002388}{0.117}$ $= 0.0204 \text{ L}$	4

Unit 2B, Chemical Reactions Revisited

11	A sample of zinc dust contains zinc oxide as an impurity. When 10.0 g of this impure zinc dust reacted with excess acid, 2.40 L of hydrogen gas at 25°C and 101.3 kPa pressure formed. Only the zinc reacts with the acid to form hydrogen gas: $\text{Zn(s)} + 2\text{H}^+(\text{aq}) \rightarrow \text{Zn}^{2+}(\text{aq}) + \text{H}_2(\text{g})$ If the molar volume of gas at 25°C and 101.3 kPa pressure is 24.5 L, what percentage of zinc was in the impure dust sample?	$n(\text{H}_2) = \frac{2.40}{24.5} = 0.09796$ $n(\text{Zn}) \text{ in impure dust} = 0.09796$ $m(\text{Zn}) = 0.09796 \times 65.38 = 6.405 \text{ g}$ $\% \text{ of Zn in impure dust} = \frac{6.405}{10.0} \times 100 = 64.0\%$	4
12	Ammonium sulfate and calcium hydroxide react in accordance with the following equation: $(\text{NH}_4)_2\text{SO}_4(\text{s}) + \text{Ca}(\text{OH})_2(\text{s}) \rightarrow \text{CaSO}_4(\text{s}) + 2\text{NH}_3(\text{g}) + 2\text{H}_2\text{O}(\text{l})$ A mixture of 109 g of ammonium sulfate and 53.0 g of calcium hydroxide is gently heated, causing the reaction to occur. What mass of ammonia is produced?	$M((\text{NH}_4)_2\text{SO}_4) = (2 \times 14.01) + (8 \times 1.008) + 32.06 + (4 \times 16.00) = 132.144 \text{ g mol}^{-1}$ $M(\text{Ca}(\text{OH})_2) = 40.08 + (2 \times 16.00) + (2 \times 1.008) = 74.096 \text{ g mol}^{-1}$ $n((\text{NH}_4)_2\text{SO}_4) = \frac{109}{132.144} = 0.8249$ $n(\text{Ca}(\text{OH})_2) = \frac{53.0}{74.096} = 0.7058$ 0.8249 mol of $(\text{NH}_4)_2\text{SO}_4$ will react with 0.8249 mol of $\text{Ca}(\text{OH})_2$, but only have 0.7058 mol of $\text{Ca}(\text{OH})_2$ in the mixture That is, $\text{Ca}(\text{OH})_2$ is the limiting reagent 0.7058 mol of $\text{Ca}(\text{OH})_2$ will form $2 \times 0.7058 = 1.412$ mol of NH_3 $M(\text{NH}_3) = 14.01 + (3 \times 1.008) = 17.034 \text{ g mol}^{-1}$ $m(\text{NH}_3) = 1.412 \times 17.034 = 24.1 \text{ g}$	5

Unit 2B, Chemical Reactions Revisited

13	A 50.0 g sample of a solid copper ore, azurite, $\text{Cu}_3(\text{OH})_2(\text{CO}_3)_2$, is to be analysed in the laboratory for the percentage of copper. The sample is crushed and excess 4.00 mol L^{-1} H_2SO_4 is added. Sand and clay impurities are filtered off and the filtrate (a solution containing copper(II) sulfate) is reacted with excess 1.00 mol L^{-1} NaOH solution to precipitate copper(II) hydroxide. This was filtered off and heated in a crucible to convert it to copper(II) oxide. The mass of copper(II) oxide formed is 6.38 g. $\text{Cu}_3(\text{OH})_2(\text{CO}_3)_2(\text{s}) + 6\text{H}^+(\text{aq}) \rightarrow 3\text{Cu}^{2+}(\text{aq}) + 4\text{H}_2\text{O}(\text{l}) + 2\text{CO}_2(\text{g})$ $\text{Cu}^{2+}(\text{aq}) + 2\text{OH}^-(\text{aq}) \rightarrow \text{Cu}(\text{OH})_2(\text{s})$ $\text{Cu}(\text{OH})_2(\text{s}) \rightarrow \text{CuO}(\text{s}) + \text{H}_2\text{O}(\text{l})$ a Calculate the mass of copper in the 6.38 g of copper(II) oxide obtained. b Calculate the percentage by mass of copper in the azurite. c Calculate the minimum volume of the 4.00 mol L^{-1} H_2SO_4 solution required to react with the 50.0 g azurite sample, as described in the first equation.	
	a $M(\text{CuO}) = 63.55 + 16.00 = 79.55 \text{ g mol}^{-1}$ $n(\text{CuO}) = \frac{6.38}{79.55} = 0.08020 = n(\text{Cu})$ $m(\text{Cu}) = 0.08020 \times 63.55 = 5.10 \text{ g}$ b All the copper in the copper oxide originated from the azurite ore, so mass of copper in the ore = 5.10 g $\% \text{Cu in the azurite} = \frac{5.10}{50.0} \times 100 = 10.2\%$ c From the 3rd equation: 0.08020 mol of CuO has formed from 0.08020 mol of $\text{Cu}(\text{OH})_2$ From the 2nd equation: 0.08020 mol of $\text{Cu}(\text{OH})_2$ has formed from 0.08020 mol of Cu^{2+} From the 1st equation: 0.08020 mol of Cu^{2+} has formed from 2×0.08020 $= 0.1604 \text{ mol of H}^+$ Because 2 mol of H^+ is formed from 1 mol of H_2SO_4 $n(\text{H}_2\text{SO}_4)$ required to react with the $\text{Cu}^{2+} = \frac{1}{2} \times 0.1604 = 0.0802 \text{ mol}$ $V(\text{H}_2\text{SO}_4 \text{ solution}) = \frac{0.0802}{4.00} = 0.0200 \text{ L}$	2 1 4

Unit 2B, Chemical Reactions Revisited

14	A white crystalline solid of molecular formula $C_xH_yO_z$ was isolated from certain fruits and found to have a relative molecular mass of 180. 0.270 g of this solid was burned in air and produced 0.400 g of carbon dioxide and 0.160 g of water. Determine the molecular formula of the compound.	$n(\text{CO}_2) = \frac{0.400}{44.01} = 0.009089$ $= n(\text{C}) \text{ in compound}$ $m(\text{C}) = 0.009089 \times 12.01 = 0.1092 \text{ g}$ $n(\text{H}_2\text{O}) = \frac{0.160}{18.016} = 0.008881$ $n(\text{H}) \text{ in compound} = 2 \times 0.008881 = 0.01776$ $m(\text{H}) = 0.01776 \times 1.008 = 0.01790 \text{ g}$ $m(\text{O}) \text{ in compound}$ $= 0.270 - (0.1092 + 0.01790)$ $= 0.1429 \text{ g}$ $n(\text{O}) = \frac{0.1429}{16.00} = 0.008931$ <table border="1" data-bbox="810 869 1347 1106"> <thead> <tr> <th></th><th>C</th><th>H</th><th>O</th></tr> </thead> <tbody> <tr> <td>mol</td><td>0.009089</td><td>0.01776</td><td>0.008931</td></tr> <tr> <td>=</td><td>$\frac{0.009089}{0.008931}$</td><td>$\frac{0.01776}{0.008931}$</td><td>$\frac{0.008931}{0.008931}$</td></tr> <tr> <td>=</td><td>1.018</td><td>1.989</td><td>1</td></tr> <tr> <td>$\approx$</td><td>1</td><td>2</td><td>1</td></tr> </tbody> </table> The empirical formula is CH_2O. If molecular formula is $(\text{CH}_2\text{O})_n$ then $(12.01 + (2 \times 1.008) + 16.00) \times n = 180$ $n = \frac{180}{30.026} = 5.995 \approx 6$ Therefore the molecular formula is $(\text{CH}_2\text{O})_6$ or $\text{C}_6\text{H}_{12}\text{O}_6$		C	H	O	mol	0.009089	0.01776	0.008931	=	$\frac{0.009089}{0.008931}$	$\frac{0.01776}{0.008931}$	$\frac{0.008931}{0.008931}$	=	1.018	1.989	1	$\approx$	1	2	1	9
	C	H	O																				
mol	0.009089	0.01776	0.008931																				
=	$\frac{0.009089}{0.008931}$	$\frac{0.01776}{0.008931}$	$\frac{0.008931}{0.008931}$																				
=	1.018	1.989	1																				
$\approx$	1	2	1																				

Chemistry for WA 1 TEST 7

Unit 2B, Acids and Bases

Name: _____ Class: _____ Date: _____

Instructions: Write answers in the right-hand column.

Score: _____ / 33 marks

Section A—Multiple choice (10 marks)

1	According to the Brønsted–Lowry theory of acids and bases, an acid is a substance that: A produces hydrogen ions in aqueous solution B produces a solution that turns litmus blue C donates a proton to a base D reacts with metals to form hydrogen gas.	C	1
2	In which of the following reactions is the underlined species acting as an acid? A $\text{H}_3\text{O}^+(\text{aq}) + \underline{\text{HPO}_4^{2-}(\text{aq})} \rightarrow \text{H}_2\text{O}(\text{l}) + \text{H}_2\text{PO}_4^-(\text{aq})$ B $\underline{\text{H}_2\text{O}(\text{l})} + \text{HCO}_3^-(\text{aq}) \rightarrow \text{H}_3\text{O}^+(\text{aq}) + \text{CO}_3^{2-}(\text{aq})$ C $\underline{\text{H}_2\text{O}(\text{l})} + \text{PH}_3(\text{g}) \rightarrow \text{PH}_4^+(\text{aq}) + \text{OH}^-(\text{aq})$ D $\text{H}_3\text{O}^+(\text{aq}) + \underline{\text{HS}^-(\text{aq})} \rightarrow \text{H}_2\text{O}(\text{l}) + \text{H}_2\text{S}(\text{aq})$	C	1
3	The carbonation ion, in sodium carbonate, is an example of a weak base. From this one could assume that: A sodium carbonate is not very soluble in water B there are relatively few hydroxide ions in the solution C the carbonate ion does not react with water molecules D blue litmus paper would turn faintly red when placed in the sodium carbonate solution.	B	1
4	The sourness of a substance is a reasonable guide to its acidity – sourness increases with increasing acidity. If solution A were to be more sour than solution B, then it is likely that : A solution A is at a higher pH than solution B B solution A has more $\text{OH}^-(\text{aq})$ ions in solution than solution B C solution A has a higher concentration of $\text{H}_3\text{O}^+(\text{aq})$ ions than solution B D solution B is a stronger acid than solution A.	C	1
5	Given three unlabelled beakers containing aqueous solutions of zinc chloride, sodium carbonate and barium nitrate, which ONE of the following reagents could be used to identify each of the solutions? A dilute nitric acid B dilute sulfuric acid C dilute hydrochloric acid D sodium chloride solution.	B	1

Unit 2B, Acids and Bases

6	Sulfuric acid is said to be a stronger acid than acetic acid because: A sulfuric acid has two ionisable hydrogen atoms and acetic acid has only one ionisable hydrogen atom B sulfuric acid ionises more completely than acetic acid in aqueous solution C aqueous sulfuric acid conducts electricity but aqueous acetic acid does not D the pH of a sulfuric acid solution is greater than that of an acetic acid solution of the same concentration.	B	1
7	A gas is bubbled through pure water, causing the pH of the water to change from 7 to 10. Of the following, the gas is most likely to be: A CO₂ B HBr C NH₃ D NO₂	D	1
8	Which group contains only acidic oxides? A CO₂, CaO, SiO₂ B P₄O₁₀, CO₂, SO₂ C K₂O, SO₂, SO₃ D H₂O, CO₂, Cl₂O	A	1
9	Excess hydrochloric acid was added to a mixture of sodium carbonate and sodium hydrogencarbonate. 1.0 mol of carbon dioxide was collected. The mixture could have contained: A 0.25 mol NaHCO₃ and 0.20 mol Na₂CO₃ B 0.50 mol NaHCO₃ and 1.00 mol Na₂CO₃ C 0.50 mol NaHCO₃ and 0.50 mol Na₂CO₃ D 0.50 mol NaHCO₃ and 0.25 mol Na₂CO₃	C	1
10	Which one of the following statements about the pH of aqueous solutions at 25⁰C is false? A The pH of a 0.1 mol L⁻¹ solution of NaCl is equal to 7. B The pH of a 0.1 mol L⁻¹ solution of NaHCO₃ is less than 7. C The pH of a 0.1 mol L⁻¹ solution of H₂SO₄ is less than the pH of a 0.1 mol L⁻¹ solution of HCl D The pH of a 0.1 mol L⁻¹ solution of NaOH is greater than the pH of a 0.1 mol L⁻¹ solution of NH₃.	B	1

Unit 2B, Acids and Bases

Section B—Written answers (23 marks)

11	Write balanced equations (ionic where appropriate) to represent the following reactions: a dilute nitric acid is added to solid magnesium hydroxide b carbon dioxide is bubbled through limewater (solution of calcium hydroxide) c a solution of sulfuric acid is added to calcium carbonate solid d lead (II) oxide solid and dilute nitric acid are mixed e hydrogen chloride gas and ammonia gas are allowed to mix f some small pieces of calcium are added to dilute hydrochloric acid.	a $2\text{H}^+(\text{aq}) + \text{Mg}(\text{OH})_2(\text{s}) \rightarrow 2\text{H}_2\text{O}(\text{l}) + \text{Mg}^{2+}(\text{aq})$ b $\text{CO}_2(\text{g}) + \text{Ca}^{2+}(\text{aq}) + 2\text{OH}^-(\text{aq}) \rightarrow \text{CaCO}_3(\text{s}) + \text{H}_2\text{O}(\text{l})$ c $2\text{H}^+(\text{aq}) + \text{CaCO}_3(\text{s}) \rightarrow \text{CO}_2(\text{g}) + \text{H}_2\text{O}(\text{l}) + \text{Ca}^{2+}(\text{aq})$ d $\text{PbO}(\text{s}) + 2\text{H}^+(\text{aq}) \rightarrow \text{Pb}^{2+}(\text{aq}) + \text{H}_2\text{O}(\text{l})$ e $\text{HCl}(\text{g}) + \text{NH}_3(\text{g}) \rightarrow \text{NH}_4\text{Cl}(\text{s})$ f $\text{Ca}(\text{s}) + 2\text{H}^+(\text{aq}) \rightarrow \text{H}_2(\text{g}) + \text{Ca}^{2+}(\text{aq})$	2
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Unit 2B, Acids and Bases

12	Give the observations for the following reactions: a When an excess of 0.1 mol L^{-1} sulfuric acid is added to some copper carbonate, carbon dioxide, water and a solution of copper sulfate are formed. b A solution of iron(III) chloride and water are formed when an iron nail is heated in 4.0 mol L^{-1} hydrochloric acid.	a When a colourless liquid is added to a green (blue) solid, a colourless odourless gas and a blue solution are formed. b The silver-coloured solid reacts with the colourless liquid to form an orange coloured solution.	3
13	Use the Brønsted–Lowry theory of acids and bases to answer the following questions: a Write an equation for the reaction of ammonia (NH_3) with water. Is the resulting solution acidic, basic or neutral? b Write an equation for the ionisation of water. Is the resulting solution acidic, basic or neutral? c Write an equation for the reaction of hydrogen fluoride (HF) with water. Is the resulting solution acidic, basic or neutral?	a $\text{NH}_3(\text{aq}) + \text{H}_2\text{O}(\text{l}) \rightarrow \text{NH}_4^+(\text{aq}) + \text{OH}^-(\text{aq})$ The solution is basic. b $2\text{H}_2\text{O}(\text{l}) \rightarrow \text{H}_3\text{O}^+(\text{aq}) + \text{OH}^-(\text{aq})$ The solution is neutral. c $\text{HF}(\text{aq}) + \text{H}_2\text{O}(\text{l}) \rightarrow \text{H}_3\text{O}^+(\text{aq}) + \text{F}^-(\text{aq})$ The solution is acidic.	3

Unit 2B, Acids and Bases

14	When dilute sulfuric acid is added to substance A (a solid), there is a vigorous effervescence and the gas explodes when ignited in air. When the remaining solution is concentrated, colourless crystals separate. A confused student is told that these crystals are either magnesium oxide, zinc sulfate, copper sulfate or calcium oxide. a What are the crystals? b Identify A c Write an ionic equation to show the reaction of A with dilute sulfuric acid.	a zinc sulphate b zinc c $\text{Zn(s)} + 2\text{H}^+(\text{aq}) \rightarrow \text{H}_2(\text{g}) + \text{Zn}^{2+}(\text{aq})$	2
15	a Nitrous acid, HNO_2, is a weak acid. With the aid of an equation, describe what this means. b Carbonic acid, H_2CO_3, is an example of a polyprotic acid. With the aid of equations, describe what this means.	a Weak acids only partially ionise in water. $\text{HNO}_2(\text{aq}) + \text{H}_2\text{O(l)} \rightarrow \text{NO}_2^-(\text{aq}) + \text{H}_3\text{O}^+(\text{l})$ This means that the solution will contain many HNO_2 molecules and not many H_3O^+ and NO_2^- ions. b A polyprotic acid can donate more than one hydrogen per molecule. For example, carbonic acid is a diprotic acid – it can donate 2 hydrogen ions: $\text{H}_2\text{CO}_3(\text{aq}) + \text{H}_2\text{O(l)} \rightarrow \text{HCO}_3^-(\text{aq}) + \text{H}_3\text{O}^+(\text{aq})$ followed by $\text{HCO}_3^-(\text{aq}) + \text{H}_2\text{O(l)} \rightarrow \text{CO}_3^{2-}(\text{aq}) + \text{H}_3\text{O}^+(\text{aq})$	2
16	Explain the following observations: a A solution of HCl(g) in water reacts vigorously with magnesium. b A solution of HCl(g) in dry ether ($\text{C}_4\text{H}_{10}\text{O}$) will not react with magnesium.	a When HCl dissolves in water, because it is a strong acid, it completely ionises to form many hydrogen ions ($\text{H}_3\text{O}^+(\text{aq})$). These hydrogen ions then react with the magnesium: $\text{HCl(g)} + \text{H}_2\text{O(aq)} \rightarrow \text{Cl}^-(\text{aq}) + \text{H}_3\text{O}^+(\text{aq})$ $\text{Mg(s)} + 2\text{H}^+(\text{aq}) \rightarrow \text{Mg}^{2+}(\text{aq}) + \text{H}_2(\text{g})$ b When HCl dissolves in ether, it does not ionise. As a result there are no hydrogen ions to react with the magnesium.	5

Unit 2B, Acids and Bases

17	a Explain why when 20 mL of a 0.1 mol L⁻¹ HCl solution is added slowly to 20 mL of a 0.1 mol L⁻¹ KOH solution, the pH of the mixture decreases from 13 to 7. b Explain why if an extra 20 mL of the 0.1 mol L⁻¹ HCl solution were to be added to the mixture formed in part a, the pH would continue to decrease to about 1.5.	a The KOH solution is basic and because there are more OH⁻ ions than H⁺ ions, the pH is greater than 7. When acid is added to this solution, the acid (H⁺) reacts with the base (OH⁻), and the concentration of OH⁻ decreases. This results in the pH decreasing: $\text{H}^+(\text{aq}) + \text{OH}^- \rightarrow \text{H}_2\text{O}(\text{l})$ (This is a neutralisation reaction) Because the amount in mole of H⁺ and OH⁻ will finally be mixed, all the OH⁻ ions will react with the H⁺ ions to form water. The solution formed will therefore be neutral (Na⁺ and Cl⁻ ions are neutral). b When extra HCl is added to the neutral mixture, more H⁺ ions are being added to the mixture. This results in a concentration of H⁺ in the mixture and as a consequence, the pH decreases (i.e. the solution becomes more acidic)	6
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Chemistry for WA 1 TEST 8

Unit 2B, Oxidation and Reduction Chemistry

Name: _____ Class: _____ Date: _____

Instructions: Write answers in the right-hand column.

Score: _____ / 30 marks

Section A—Multiple choice (10 marks)

1	Which one of the following statements contains two correct descriptions of oxidation? A addition of electrons, decrease in oxidation number B removal of electrons, decrease in oxidation number C addition of electrons, increase in oxidation number D removal of electrons, increase in oxidation number.	D	1
2	Which of the following statements is false? A The reactant from which electrons are taken is reduced. B Oxidation involves an increase in oxidation state. C A reductant undergoes oxidation. D In reduction half equations, the electrons appear on the left hand side (reactant side) of the equation.	A	1
3	Which one of the following is the <u>best</u> definition of an oxidant? An oxidant: A always contains oxygen B causes the oxidation of another substance C is oxidised during reactions D donates electrons to another substance.	B	1
4	In the following reaction: $\text{H}_2\text{S}(\text{g}) + \text{Br}_2(\text{aq}) \rightarrow \text{S}(\text{s}) + 2\text{H}^+(\text{aq}) + 2\text{Br}^-(\text{aq})$ A H_2S is oxidised B Br_2 is the reductant C H_2S has undergone reduction D neither H_2S nor Br_2 have been reduced or oxidised.	A	1
5	In which of the following species is vanadium exhibiting the highest oxidation number? A VO^{2+} B NH_4VO_3 C $\text{V}_2(\text{SO}_4)_3$ D VBr_4	B	1
6	Which of the following reactions is a redox reaction? A $\text{CaCO}_3(\text{s}) \rightarrow \text{CaO}(\text{s}) + \text{CO}_2(\text{g})$ B $\text{Pb}^{2+}(\text{aq}) + 2\text{Br}^-(\text{aq}) \rightarrow \text{PbBr}_2(\text{s})$ C $2\text{H}^+(\text{aq}) + \text{HCO}_3^-(\text{aq}) \rightarrow \text{CO}_2(\text{g}) + \text{H}_2\text{O}(\text{l})$ D $\text{MnO}_4^-(\text{aq}) + 5\text{Fe}^{2+}(\text{aq}) + 8\text{H}^+(\text{aq}) \rightarrow \text{Mn}^{2+}(\text{aq}) + 5\text{Fe}^{3+}(\text{aq}) + 4\text{H}_2\text{O}(\text{l})$	D	1

Unit 2B, Oxidation and Reduction Chemistry

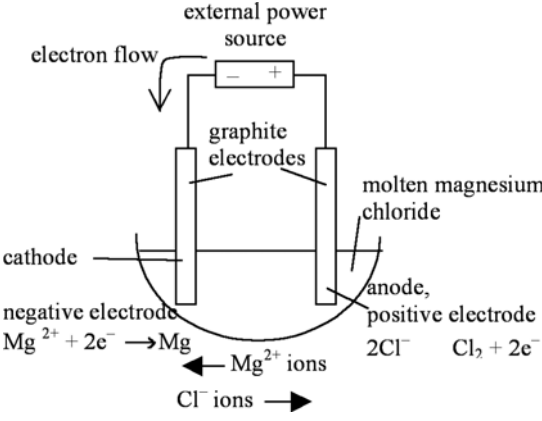
7	Which equation shows the disproportionation of chlorine (i.e. chlorine acts as both the oxidant and the reductant)? A $\text{Cl}_2(\text{aq}) + \text{H}_2\text{O}(\text{l}) \rightarrow \text{H}^+(\text{aq}) + \text{Cl}^-(\text{aq}) + \text{HClO}(\text{aq})$ B $\text{Cl}_2(\text{g}) + 2\text{e}^- \rightarrow 2\text{Cl}^-(\text{aq})$ C $\text{Cl}_2(\text{g}) + \text{H}_2(\text{g}) \rightarrow 2\text{HCl}(\text{g})$ D $\text{Cl}_2(\text{aq}) + 2\text{H}_2\text{O}(\text{l}) \rightarrow 2\text{HClO}(\text{aq}) + 2\text{H}^+(\text{aq}) + 2\text{e}^-$	A	1
8	During the electrolysis of a molten salt, which one of the following statements is false? A Reduction takes place at the cathode. B The positive electrode is the anode. C Electrons flow from the negative terminal of the power source into the cathode. D The positive ions in the electrolyte flow towards the anode.	D	1
9	A series of reactions is carried out between the oxidants A^{2+} , B^{2+} , C^{2+} , D^{2+} and the corresponding reductants A, B, C, D. It is found that: i C^{2+} does not react with B ii A^{2+} does not react with B iii D^{2+} does not react with A iv C^{2+} reacts with A The strongest oxidant (the substance that 'wants' to gain electrons the most) in the group is: A A^{2+} B B^{2+} C C^{2+} D D^{2+}	B	1
10	An electric current is passed through molten calcium oxide, using inert electrodes. Which of the following statements about this electrolysis process is not correct? A Oxygen gas is produced at the anode. B One atom of calcium is deposited for each molecule of oxygen liberated. C The oxide ions flow towards the negative electrode. D Calcium ions are reduced during the electrolysis process.	C	1

Unit 2B, Oxidation and Reduction Chemistry

Section B—Written answers (20 marks)

11	Identify the reactant being oxidised and the reactant being reduced in the following reactions: a $\text{Zn(s)} + 4\text{H}^+(\text{aq}) + 2\text{NO}_3^-(\text{aq}) \rightarrow \text{Zn}^{2+}(\text{aq}) + 2\text{NO}_2(\text{g}) + 2\text{H}_2\text{O(l)}$ b $2\text{K(s)} + 2\text{H}_2\text{O(l)} \rightarrow 2\text{K}^+(\text{aq}) + 2\text{OH}^-(\text{aq}) + \text{H}_2(\text{g})$ c $\text{Fe(s)} + \text{Br}_2(\text{l}) \rightarrow \text{FeBr}_2(\text{s})$ d $\text{SO}_2(\text{g}) + 2\text{H}_2\text{S(g)} \rightarrow 2\text{H}_2\text{O(l)} + 3\text{S(s)}$ a Zn oxidised, NO_3^- reduced b K oxidised, H_2O reduced c Fe oxidised, Br_2 reduced d SO_2 reduced, H_2S oxidised	2
12	Identify the first reactant in the following reactions as an oxidant or a reductant a $\text{H}_2\text{O}_2(\text{aq}) + \text{NO}_2^-(\text{aq}) \rightarrow \text{H}_2\text{O(l)} + \text{NO}_3^-(\text{aq})$ b $3\text{SO}_2(\text{g}) + \text{Cr}_2\text{O}_7^{2-}(\text{aq}) + 2\text{H}^+(\text{aq}) \rightarrow 3\text{SO}_4^{2-}(\text{aq}) + 2\text{Cr}^{3+}(\text{aq}) + \text{H}_2\text{O(l)}$ a H_2O_2 – oxidant b SO_2 – reductant	3
13	Give the oxidation number of each of the atoms in the following substances: NH_3, NaNO_2, CaH_2, $\text{Fe}_2(\text{CO}_3)_3$ NH_3: N -3, H +1 NaNO_2: Na +1, N +3, O -2 CaH_2: Ca +2, H -1 $\text{Fe}_2(\text{CO}_3)_3$: Fe +3, C +4, O -2	2

Unit 2B, Oxidation and Reduction Chemistry

14	For each of the following reactions, write the oxidation and the reduction half equations and the overall equation. a when chlorine gas is bubbled through a solution of sodium bromide, the solution turns an orange/brown colour. b white shiny crystals grow on a piece of magnesium when it is placed in a solution of silver nitrate. c when pieces of aluminium are heated with hydrochloric acid, bubbles of a colourless gas are formed.	a oxidation: $2\text{Br}^-(\text{aq}) \rightarrow \text{Br}_2(\text{aq}) + 2\text{e}^-$ reduction: $\text{Cl}_2(\text{g}) + 2\text{e}^- \rightarrow 2\text{Cl}^-(\text{aq})$ overall: $2\text{Br}^-(\text{aq}) + \text{Cl}_2(\text{g}) \rightarrow \text{Br}_2(\text{aq}) + 2\text{Cl}^-(\text{aq})$ b oxidation: $\text{Mg}(\text{s}) \rightarrow \text{Mg}^{2+}(\text{aq}) + 2\text{e}^-$ reduction: $\text{Ag}^+(\text{aq}) + \text{e}^- \rightarrow \text{Ag}(\text{s})$ overall: $2\text{Ag}^+(\text{aq}) + \text{Mg}(\text{s}) \rightarrow 2\text{Ag}(\text{s}) + \text{Mg}^{2+}(\text{aq})$ c oxidation: $\text{Al}(\text{s}) \rightarrow \text{Al}^{3+}(\text{aq}) + 3\text{e}^-$ reduction: $2\text{H}^+(\text{aq}) + 2\text{e}^- \rightarrow \text{H}_2(\text{g})$ overall: $2\text{Al}(\text{s}) + 6\text{H}^+(\text{aq}) \rightarrow 2\text{Al}^{3+}(\text{aq}) + 3\text{H}_2(\text{g})$	2
15	Draw a labelled diagram to show how you would use the process of electrolysis to produce magnesium from magnesium chloride. In your diagram, indicate: a whether a power source is needed b the electrolyte to be used c the anode and the cathode d the direction of electron and ion flow e the positive and negative electrodes f the half equations for the reactions occurring at each electrode.		5

Unit 2B, Oxidation and Reduction Chemistry

12	An electric current is passed through molten sodium bromide, using graphite electrodes. a What particles conduct the electricity in the wires in the external part of the circuit? b What particles act as the charge carriers in the internal part of the circuit, and in which direction do they flow? c What substance is produced at the cathode d Give the half equation that occurs at the anode. Is this a reduction or an oxidation reaction? e Explain in terms of its bonding and structure, why graphite can be used as an electrode.	a the delocalised electrons b positive and negative ions - Na^+ ions move towards the cathode (negative electrode) - Br^- ions move towards the anode (positive electrode) c sodium metal d $2\text{Br}^-(\text{l}) \rightarrow \text{Br}_2(\text{g}) + 2\text{e}^-$ - oxidation e Graphite is a covalent network substance made up of layers of carbon atoms covalently bonded together. Each carbon atom is bonded to three other atoms, leaving one electron per atom that is free to move between the layers. These delocalised electrons are able to move under the influence of an electric field. That is, graphite is an excellent conductor of electricity. Because of the strong bonds holding the carbon atoms in the layers, graphite has a high melting point and so remains a solid when placed in the molten ionic salt. Graphite is also reasonably unreactive and so does not react with the chemicals in the cell.	5
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Unit 2B, Organic Chemistry

Name: _____ Class: _____ Date: _____

Instructions: Write answers in the right-hand column.

Score: _____ / 42 marks

Section A—Multiple choice (10 marks)

1	How many structural isomers will have the formula C_4C_9Cl ? A 2 B 3 C 4 D 5	C	1
2	A compound has the formula C_7H_{14} . Which one of the following could it be? A A straight chain alkane B An alkene with one double bond C A cycloalkene D An aromatic hydrocarbon (a hydrocarbon containing a benzene ring)	B	1
3	When an organic compound was treated with chlorine, Cl_2 , in the presence of ultraviolet light, the major reaction product was chlorocyclohexane. Which one of the following was the original organic compound? A benzene  B cyclohexane  C cyclohexene  D hexane $CH_3CH_2CH_2CH_2CH_2CH_3$	B	1
4	A particular organic compound has the formula of $\begin{array}{c} CH_3 \\ \\ CH_3-CH-CH-CH_3 \\ \\ CH_3 \end{array}$ Which of the following statements about this compound is true? A The name of the compound is 2,3-dimethylhexane. B When mixed with a solution of bromine the orange colour of the bromine will rapidly fade. C It will burn in an excess of air to form carbon dioxide and water. D It will form geometric isomers.	C	1

Unit 2B, Organic Chemistry

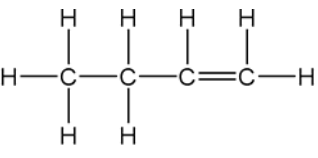
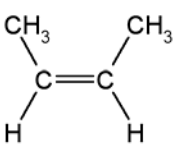
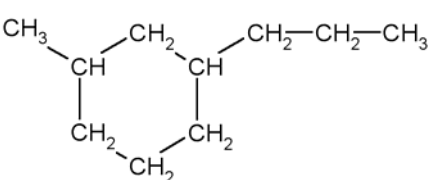
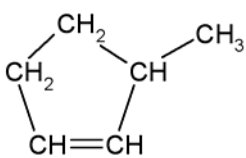
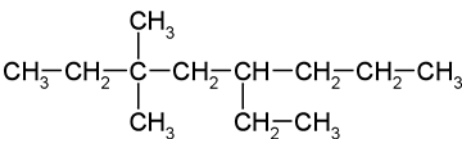
5	Each organic compound has at least one correct name. How many of the following names could NOT be correct? • 1,2-dimethylpropane • but-3-ene • 2-methylbut-2-ene • 1,1,1-trimethylcyclopentane • <i>trans</i>-hex-2-ene A 1 B 2 C 3 D 4	C	1
6	Which formula represents two substances that are geometric (<i>cis/trans</i>) isomers? A C₄H₁₀ B C₃H₆ C CH₂=CHCH₂CH₃ D CH₃CH=CHCH₂CH₃	D	1
7	Which of the following pairs of compounds are pairs of isomers? i heptane and 2-methylheptane ii pentane and cyclopentane iii hex-2-ene and cyclohexane iv 2-methylhexane and ethylpentane A iii and iv B i and ii C ii and iii D i and iv	A	1
8	If an equal amount, in mole, of each of the following is burned in excess oxygen, which will give the greatest mass of CO₂? A C₂H₄ B C₂H₆ C C₃H₈ D C₄H₈	D	1
9	Which of the following statements best explains the stability of the benzene ring? A Benzene is an unsaturated hydrocarbon. B The benzene ring structure contains three double bonds. C Six delocalised electrons are present in the ring structure of benzene. D Cyclic hydrocarbons are more stable than straight chain hydrocarbons.	C	1

Unit 2B, Organic Chemistry

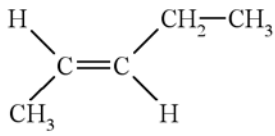
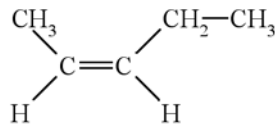
10	Which of the following reactions are substitution reactions? i $\text{CH}_4 + \text{Cl}_2 \rightarrow \text{CH}_3\text{Cl} + \text{HCl}$ ii $\text{CH}_2=\text{CH}_2 + \text{Br}_2 \rightarrow \text{CH}_2\text{Br}-\text{CH}_2\text{Br}$ iii $\text{CH}_2=\text{CH}_2 + \text{H}_2 \rightarrow \text{CH}_3-\text{CH}_3$ iv $\text{C}_6\text{H}_6 + \text{Cl}_2 \rightarrow \text{C}_6\text{H}_5\text{Cl} + \text{HCl}$ A i and iv B ii and iii C i, iii and iv D ii, iii and iv	A	1
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Unit 2B, Organic Chemistry

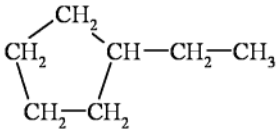
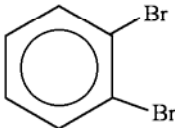
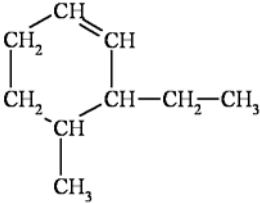
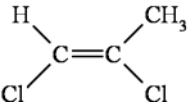
Section B—Written answers (32 marks)

11	Give the IUPAC systematic name for each of the following compounds: a  b  c  d  e 	a but-1-ene b <i>cis</i>-but-2-ene c 1-methyl-3-propylcyclohexane d 3-methylcyclopentene e 5-ethyl-3,3-dimethyloctane	5
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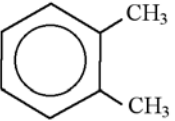
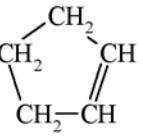
Unit 2B, Organic Chemistry

12	Give the structural formulas and names of the non-cyclic structural and geometrical isomers with the molecular formula of C_5H_{10} .	$CH_2=CH-CH_2-CH_2-CH_3$ pent-1-ene  $trans$-pent-2-ene  cis-pent-2-ene $CH_2=C(CH_3)-CH_2-CH_3$ 2-methylbut-1-ene $CH_3-C(CH_3)=CH-CH_3$ 2-methylbut-2-ene $CH_3-CH(CH_3)-CH=CH_2$ 3-methylbut-1-ene	6
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Unit 2B, Organic Chemistry

13	Give the structural formulas of the following compounds: a 3-methyloctane b 2,2-dimethylhex-3-ene c ethylcyclopentane d 1,2-dibromobenzene e 3-ethyl-4-methylcyclohexene f <i>cis</i>-1,2-dichloropropene	a $\text{CH}_3\text{—CH}_2\text{—}\underset{\text{CH}_3}{\text{CH}}\text{—CH}_2\text{—CH}_2\text{—CH}_2\text{—CH}_2\text{—CH}_3$ b $\text{CH}_3\text{—}\overset{\text{CH}_3}{\underset{\text{CH}_3}{\text{C}}}\text{—CH=CH—CH}_2\text{—CH}_3$ c  d  e  f 	6
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Unit 2B, Organic Chemistry

14	Write balanced equations for the following reactions. Use structural formula for the organic compounds: a Propene and chlorine gases are mixed and allowed to react, in the absence of ultraviolet light. b Dimethylpropane is mixed with bromine in a 1:1 mole ratio and placed in sunlight for several hours. c Cyclopentene is reacted with hydrogen, in the presence of a catalyst. d Hexane is burnt in the presence of an excess of air. e Benzene reacts with bromine in the presence of a catalyst.	a $\text{CH}_2=\text{CH}-\text{CH}_3 + \text{Cl}_2 \longrightarrow \begin{array}{c} \text{CH}_2-\text{CH}-\text{CH}_3 \\ \quad \\ \text{Cl} \quad \text{Cl} \end{array}$ b $\begin{array}{c} \text{CH}_3 \\ \\ \text{CH}_3-\text{C}-\text{CH}_3 \\ \\ \text{CH}_3 \end{array} + \text{Br}_2 \longrightarrow \begin{array}{c} \text{CH}_3 \\ \\ \text{CH}_3-\text{C}-\text{CH}_2\text{Br} \\ \\ \text{CH}_3 \end{array} + \text{HBr}$ c $\begin{array}{c} \text{CH} \\ / \quad \backslash \\ \text{CH}_2 \quad \text{CH} \\ \quad \\ \text{CH}_2-\text{CH}_2 \end{array} + \text{H}_2 \longrightarrow \begin{array}{c} \text{CH}_2 \\ / \quad \backslash \\ \text{CH}_2 \quad \text{CH}_2 \\ \quad \\ \text{CH}_2-\text{CH}_2 \end{array}$ d $2\text{CH}_3\text{CH}_2\text{CH}_2\text{CH}_2\text{CH}_2\text{CH}_3 + 19\text{O}_2 \longrightarrow 12\text{CO}_2 + 14\text{H}_2\text{O}$ e  + Br₂  + HBr	10
15	Identify by structural formula an example of each of the following: a a branched chain alkane containing five carbon atoms per molecule b an alkene other than pent-1-ene containing five carbon atoms per molecule c an aromatic compound containing eight carbon atoms per molecule d a cyclic compound that would rapidly decolourise a bromine solution e a compound containing eight hydrogen atoms per molecule that would undergo an addition reaction with chlorine.	There are other possible answers apart from the ones given below. a $\begin{array}{c} \text{CH}_3-\text{CH}-\text{CH}_2-\text{CH}_3 \\ \\ \text{CH}_3 \end{array}$ b $\text{CH}_3-\text{CH}=\text{CH}-\text{CH}_2-\text{CH}_3$ c  d  e $\text{CH}_2=\text{CH}-\text{CH}_2-\text{CH}_3$	5